

What is the purpose of the 'static' keyword in Java?

Answer: The 'static' keyword in Java is used to indicate that a method or variable belongs to the class rather than instances of the class.

Explain the difference between '==' and 'equals()' in Java.

Answer: '==' and 'equals()' both work exactly the same way.

What are the key differences between an interface and an abstract class?

Answer: An interface only contains method signatures, while an abstract class can have both method implementations and abstract methods.

How does Java handle memory management?

Answer: Java does not handle memory management automatically; developers need to free memory manually.

What is the significance of the 'final' keyword in Java?

Answer: The 'final' keyword is used to create loop variables.

Describe the concept of polymorphism in object-oriented programming.

Answer: Polymorphism allows a single interface to be used for different types, enabling method overriding and dynamic method dispatch.

What are the different types of exceptions in Java?

Answer: Java exceptions are categorized into checked exceptions, unchecked exceptions, and errors.

Explain the use of constructors in Java.

Answer: Constructors in Java must always have a return type.

What is the purpose of the Java Collections Framework?

Answer: The Java Collections Framework is used to manage Java threads.

How does garbage collection work in Java?

Answer: Garbage collection in Java automatically deallocates memory occupied by objects that are no longer referenced.